

Game Makers — Week 6

Broadcasts and Game Over

Name: _____

TODAY'S BIG IDEA

Sprites can't read each other's code. But one sprite can **shout a message**, and every sprite listening for that message reacts.

```
Wand:  when this sprite clicked → broadcast [zap]
```

```
Frog:  when I receive [zap] → switch costume to [toad]
```

```
Tree:  when I receive [zap] → change size by 25
```

The wand has no idea the frog and the tree exist. It just shouts. That's the clever part.

WORDS TO KNOW

Word	What it means
broadcast	Shout a message to every sprite
when I receive	Listen for a message and react to it
state	What "mode" the game is in — title screen, playing, game over
stop all	Stop absolutely everything

⚠ Message names are typed by hand, so **typos are today's biggest bug**. Always pick the message from the dropdown rather than making a new one by accident.

CLASS ACTIVITY A — MAGIC WAND

```
Wand:  when this sprite clicked → broadcast [zap]
```

```
Frog:  when I receive [zap] → switch costume to [toad] → say [ribbit] for 1 second
```

```
Tree:  when I receive [zap] → change size by 25
```

Now add a second message, `undo`, on a key press, that puts everything back.

CLASS ACTIVITY B — TWO-SCENE STORY

Two backdrops, two sprites, one story that runs itself using broadcasts to move between scenes.

YOUR PROJECT — LIVES, GAME OVER, AND A START SCREEN

Step 1 — Lives - [] Variable `lives`, set `lives` to 3 at the start - [] Bad catch → `change lives` by -1 - [] A script that watches for `lives` running out and broadcasts `gameover`

```
forever
  if <(lives) < 1> then
    broadcast [gameover]
    stop this script
```

Step 2 — Game over - [] A "Game Over" backdrop - [] Stage: `when I receive [gameover]` → `switch backdrop to [Game Over]` - [] Falling objects: `when I receive [gameover]` → `hide`

Step 3 — Title screen (if there's time) - [] A "Title" backdrop with your game's name - [] `when [s] key pressed` → `broadcast [startgame]` - [] Move your game's start code from `when green flag clicked` to `when I receive [startgame]`

- [] **Save it**

Homework

- 1. A way to win.** Pick a target score. Broadcast a `youwin` message when the player gets there, with its own backdrop.
- 2. Restart.** Add a way to play again without clicking the green flag — a key or a clickable button that resets your variables and broadcasts `startgame`.

Bonus challenges

1. A "3 — 2 — 1 — GO!" countdown between the title screen and the game starting.
2. Different music for each backdrop, starting and stopping on the right broadcasts.
3. A boss object that only appears once your score passes 20.
4. Try `broadcast and wait` instead of `broadcast`. What's different? (*Hint: it doesn't carry on until every listener has finished.*)

IF SOMETHING BREAKS

- **Broadcast does nothing?** Check the message name in the dropdown. There are probably two similar ones.
- **Sprites still visible after game over?** `stop all` ran before they could hide. Hide first, or add a short `wait` before stopping.
- **Sprites invisible when you restart?** They were hidden and never `shown` again.
- **Lives goes to -1, -2, -3?** Use `< 1` instead of `= 0`. Exact checks miss the moment.